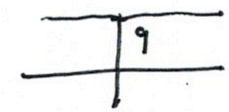
Section 3.1 The Power Rule

$y = 9$



$$\frac{d}{dx}(x^n) = n \cdot x^{n-1}$$

Ex ① $f(x) = 3 \cdot x^{3-1} + -4 \cdot x^{-4-1} + 1 \cdot 7x^{1-1} + 0 \cdot 9 \cdot x^{0-1}$

$f'(x) = 3x^2 - 4x^{-5} + 7$

$$f'(x) = 3x^2 - \frac{4}{x^5} + 7$$

② $y = \sqrt{x}$

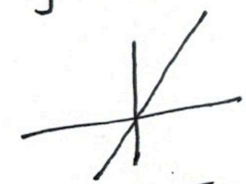
$y = \frac{1}{2} \cdot x^{1/2 - 1}$ or $\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$

$y' = \frac{1}{2} x^{-\frac{1}{2}}$

$y' = \frac{1}{2x^{1/2}}$

$y' = \frac{1}{2\sqrt{x}}$

$y = 5x$



$y' = 5$

Ex ③ $y = -2x^2 + 5x^1 + 8$

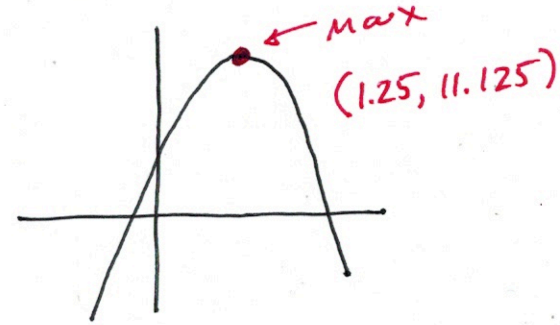
where is $y' = 0$?

$$y' = -4x + 5$$

$$0 = -4x + 5$$

$$4x = 5$$

$$x = \frac{5}{4} = 1.25$$



Ex ④ when is y' undefined?

$$y = \sqrt[5]{x-1}$$

$$y = \frac{1}{5}\cdot(x-1)^{1/5 - 1}$$

$$y' = \frac{1}{5}(x-1)^{-4/5}$$

$$y' = \frac{1}{5(x-1)^{4/5}}$$

$$5(x-1)^{4/5} = 0$$

$$\left((x-1)^{4/5}\right)^{5/4} = (0)^{5/4}$$

$$(x-1) = 0$$

$$\boxed{x = 1}$$

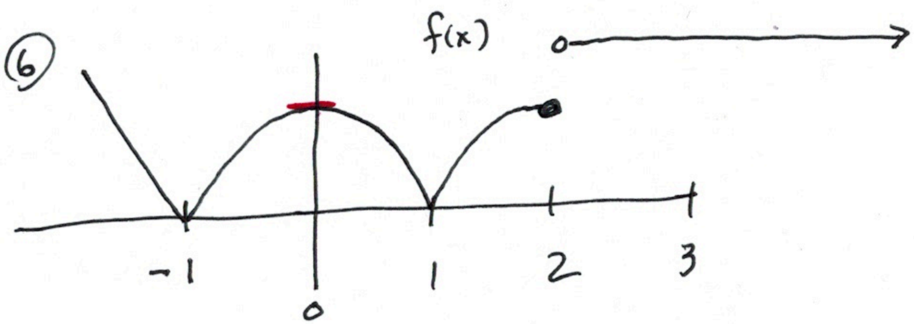
Ex ⑤ $y = \frac{1}{x}$ $\qquad \frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2}$

$y = -1 \cdot x^{-1-1}$

$y' = -1 \cdot x^{-2}$

$$\boxed{y' = -\frac{1}{x^2}}$$

Ex ⑥



A) when is $f'(x) = 0$

$(2, \infty)$ and $x = 0$

B) when is $f'(x)$ undefined?

$x = -1, \ x = 1, \ x = 2$